

Quaid Garton

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Objective

An outdoor enthusiast who loves physics too much to leave the engineering world behind. Seeking to apply a strong foundation in orbital mechanics, trajectory propagation, and simulation-based state estimation to develop guidance, navigation, and control solutions for spacecraft.

Education

University of Colorado Boulder | B.S. Aerospace Engineering with a minor in business

Professional Experience

Manufacturing Engineer - Sierra Space, Madison, WI | Nov 2022 – Dec 2023

- Analyzed and tested 3 sensor package designs for the Dream Chaser spaceplane, identifying failure modes and performance limitations
- Performed root cause analysis on failures identified during random vibration, shock, and thermal vacuum testing
- Coordinated simultaneous manufacturing and test operations for multiple sensor packages, aligning priorities across subsystem, vehicle integration, and test teams
- Kept internal manufacturing and test teams aligned on build/test status, while coordinating directly with external manufacturing and test facilities to define requirements and ensure components were built and tested to spec

Electrical Engineer – STEAM, Boulder, CO | Summer 2022

- Designed and updated PCB layouts and schematics in Altium Designer, maintaining version control with Git

Teaching Fellow – Experimental Computational Methods, Boulder, CO | Aug 2021 – May 2022

- Mentored students in modeling error propagation through a 6-DOF bottle rocket trajectory propagator
- Built Monte Carlo simulations translating measurement uncertainty into landing-location confidence intervals
- Held weekly office hours focused on coding support, debugging numerical integrators and simulation code

Whitewater Rafting Guide - Timberline Tours and Sierra Whitewater, CO and CA | May 2018 – Present

- Acted as trip leader, managing logistics, team coordination, and safety in dynamic, high-risk environments
- Instructed new crews daily on paddling techniques and routes through the rapids to ensure a safe trip

Guidance, Navigation, and Control (GNC) Related Projects

Flight Dynamics Portfolio Project – Python (Lambert Solver + Porkchop plot) | 2026

- Developed a 0-revolution Lambert Solver to compute transfer arcs with a given time-of-flight
- Built a porkchop plot workflow to evaluate launch date and time-of-flight trade spaces using Δ -v-based screening
- Verified Lambert-solver solution by numerically propagating the resulting transfer arc (e.g., Earth-Mars)
- (In progress) Developing an Extended Kalman Filter to fuse noisy measurements with propagated state estimates

Star Tracker Design – CubIST: Cubesat Integrated Star Tracker, Senior Project, CU Boulder | Fall 2021 – Spring 2022

- Designed and prototyped a cubesat-focused star tracker using COTS hardware (Raspberry Pi-based system)
- Used night sky testing to validate the attitude determination precision of the camera/software combination
- Designed and 3D printed a custom enclosure for the optical and electronic components using SolidWorks
- Presented at design reviews at many steps in the process from design to manufacture to test

3 Body Problem Analysis – NASA Astrocamp, Remote | Summer 2021

- Studied chaotic behavior and practiced targeting trajectories in the Circular Restricted 3-Body Problem

Skills

Technical skills: Matlab | Python | C++ | Git | GMAT | STK | SolidWorks | Monte Carlo Simulation | Error Propagation
Data analysis | Trajectory Design | Circular restricted 3 body problem (CR3BP) | Attitude and Orbit Control System (AOCS)

Interpersonal Skills: Strong communicator | Guest services | Team-oriented | Risk assessment | Detail Oriented